

Statistical Methods for Data Science (Spring 2026)

Mini Project #2

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SECTION 1

QUESTION 1

Reading the roadrace.csv dataset using R

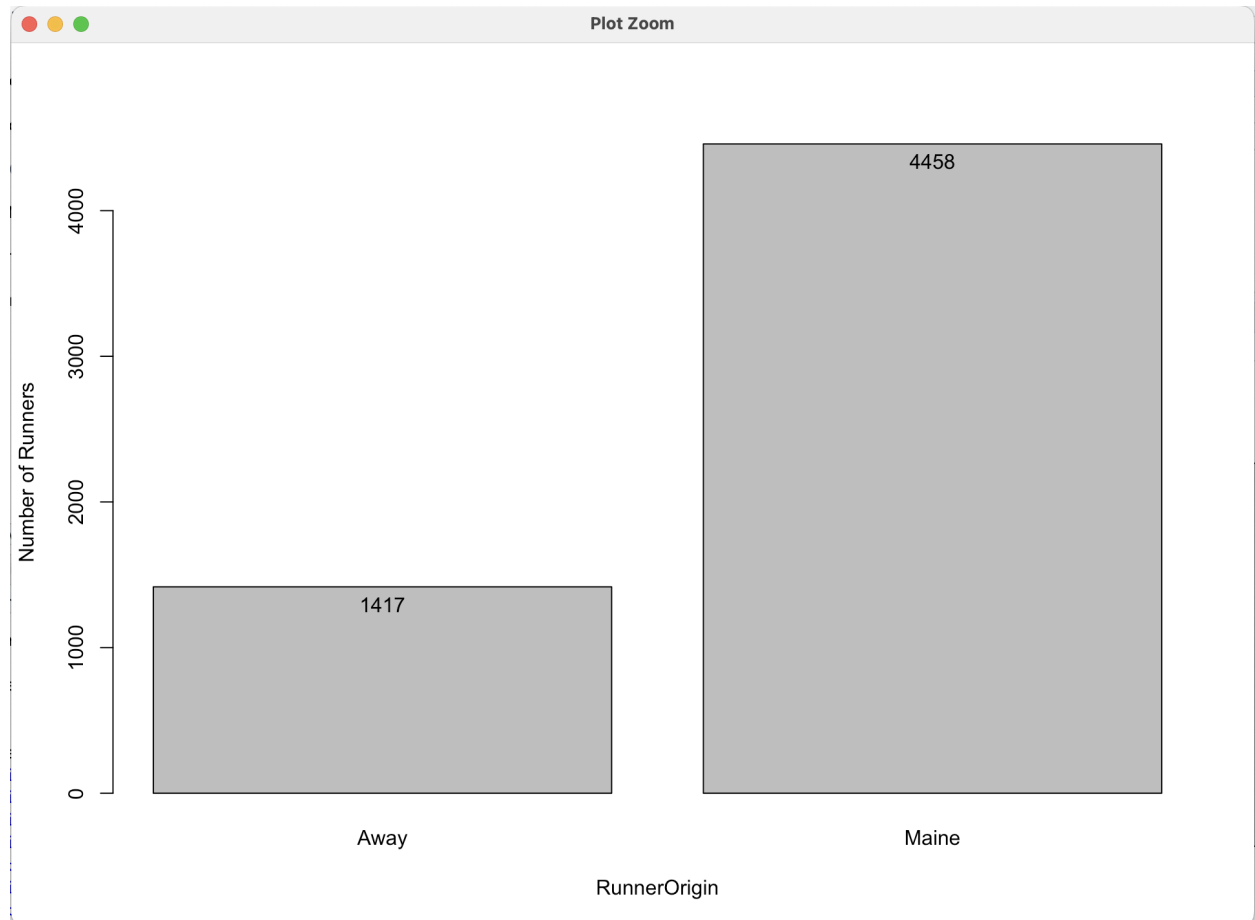
```
1 library(pacman)
2
3 pacman::p_load(rio)
4
5
6 roadrace <- import("/Users/fathimafaridamohammad/Downloads/roadrace.csv")
7
8 head(roadrace)
```

> head(roadrace)

	Place	Division	Place	Division	Entrants	Division	Age	Sex	State/Country
1	1		1		246	M2529	25	M	KENYA
2	2		1		186	M2024	21	M	KENYA
3	3		2		246	M2529	28	M	KENYA
4	4		3		246	M2529	25	M	KENYA
5	5		2		186	M2024	21	M	KENYA
6	6		3		186	M2024	22	M	KENYA

	Time (seconds)	Mile pace (seconds)	From USA	Maine	Time (minutes)
1	1666.9	269	No	Away	27.782
2	1675.1	270	No	Away	27.918
3	1679.4	271	No	Away	27.990
4	1680.1	271	No	Away	28.002
5	1691.9	273	No	Away	28.198
6	1714.1	276	No	Away	28.568

A. Bar graph of the variable Maine



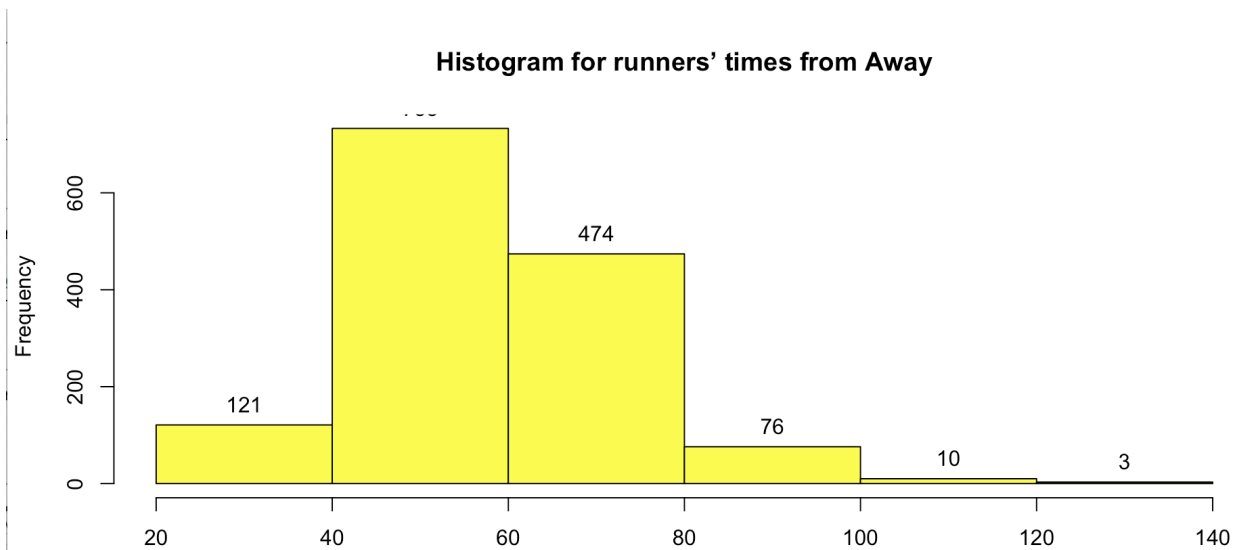
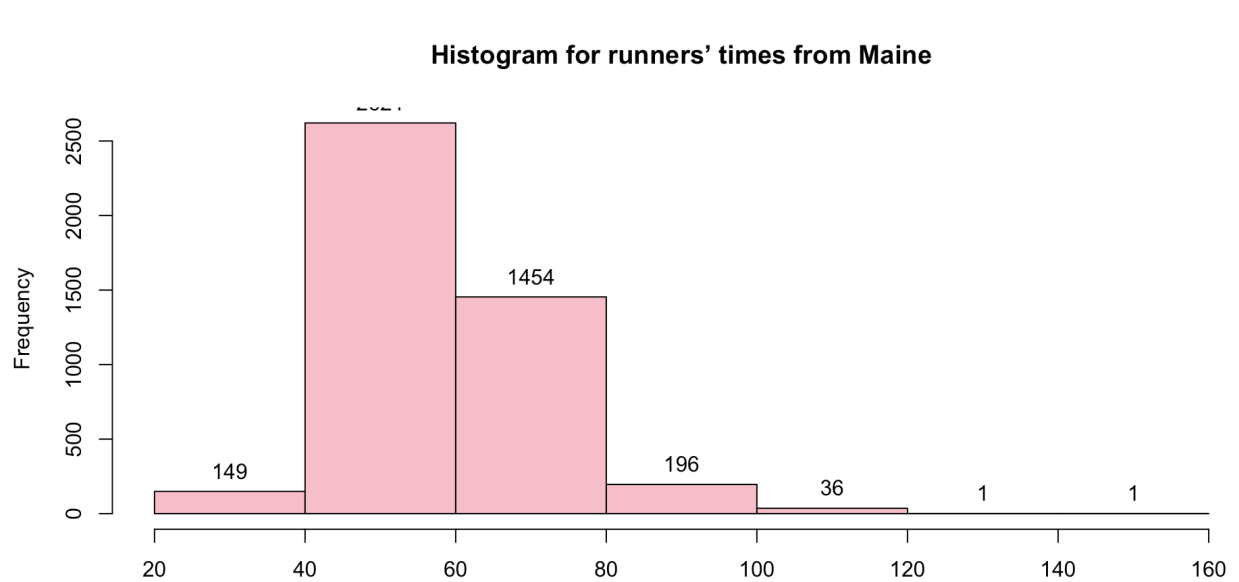
- The bar graph shows that the majority of runners in the race are from Maine. The number of runners from Maine is much larger than those from outside the state. This is supported by the below summary statistics:

```
> prop.table(MaineOrNot)
```

```
      Away      Maine  
0.2411915 0.7588085
```

- 75.88% of runners are from Maine and 24.12% of runners are from away.

B. Histograms of runners' times: Group from Maine and Group from Away



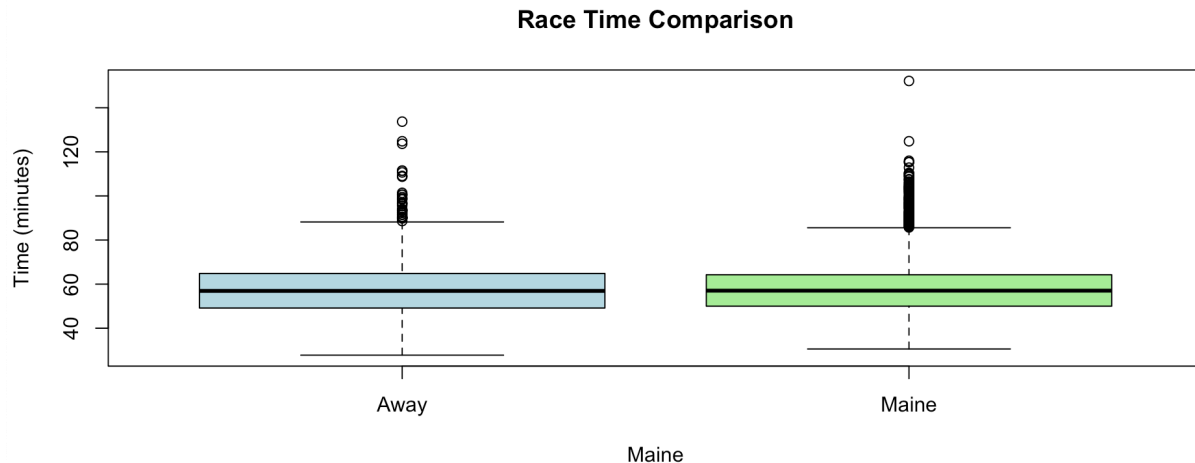
Summary statistics :

```
> summary(Maine_Runners$`Time (minutes)`)  
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   
 30.57  50.00  57.03  58.20  64.24 152.17  
> summary(Away_Runners$`Time (minutes)`)  
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   
 27.78  49.15  56.92  57.82  64.83 133.71  
> ?summary  
> sd(Maine_Runners$`Time (minutes)`)  
[1] 12.18511  
> range(Maine_Runners$`Time (minutes)`)  
[1] 30.567 152.167  
> sd(Away_Runners$`Time (minutes)`)  
[1] 13.83538  
> range(Away_Runners$`Time (minutes)`)  
[1] 27.782 133.710
```

- The histograms for both Maine and Away runners show **right-skewed distributions**, indicating that while most runners finish within a moderate time range, there are a few runners with significantly higher completion times.
- Comparing the centers of the distributions, Away runners tend to be slightly faster. This is supported by their lower mean (57.82 minutes) and lower median (56.92 minutes) compared to Maine runners, whose mean is 58.20 minutes and median is 57.03 minutes.
- In terms of variability, Away runners actually show slightly greater spread. Their standard deviation (13.84) is higher than that of Maine runners (12.19), indicating more variability in finishing times.
- Additionally, while Maine runners have a larger maximum value (152.17 minutes vs. 133.71 minutes), suggesting more extreme slow runners, the overall spread is higher for Away runners.
- **Interquartile range:**
 - Maine: $64.24 - 50.00 = 14.24$

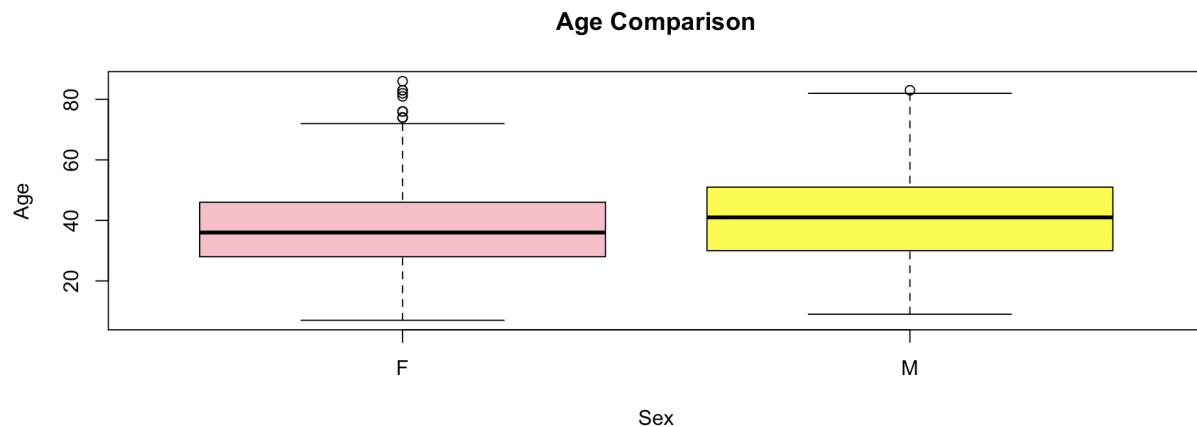
- Away: $64.83 - 49.15 = 15.68$
 $64.83 - 49.15 = 15.68$
- This confirms that Away runners have a slightly wider middle spread as well.

C. Side by side boxplots of race time for Away and Maine runners



- The two groups have very similar typical performance (mean and median nearly identical). Maine runners are very slightly slower on average because mean and median both are a bit higher, but the difference is negligible.
- Away runners have higher Standard Deviation and larger IQR compared to Maine runners. Away runners are more variable, with a wider spread of times.
- Maine runners have a larger maximum and show more extreme outliers. Away is more spread overall.
- Both groups are right-skewed, meaning most runners cluster at faster times with a tail of slower finishers. The skew appears slightly stronger for Maine due to the very large maximum.

D. Side by side boxplots for Age of Male and Female runners



Summary Statistics :

```
> summary(filter(roadrace, Sex== "M")$Age)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  9.00  30.00  41.00  40.45  51.00  83.00

> summary(filter(roadrace, Sex== "F")$Age)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  7.00  28.00  36.00  37.24  46.00  86.00

> sd(filter(roadrace, Sex=="M")$Age)
[1] 13.99289

> sd(filter(roadrace, Sex=="F")$Age)
[1] 12.26925

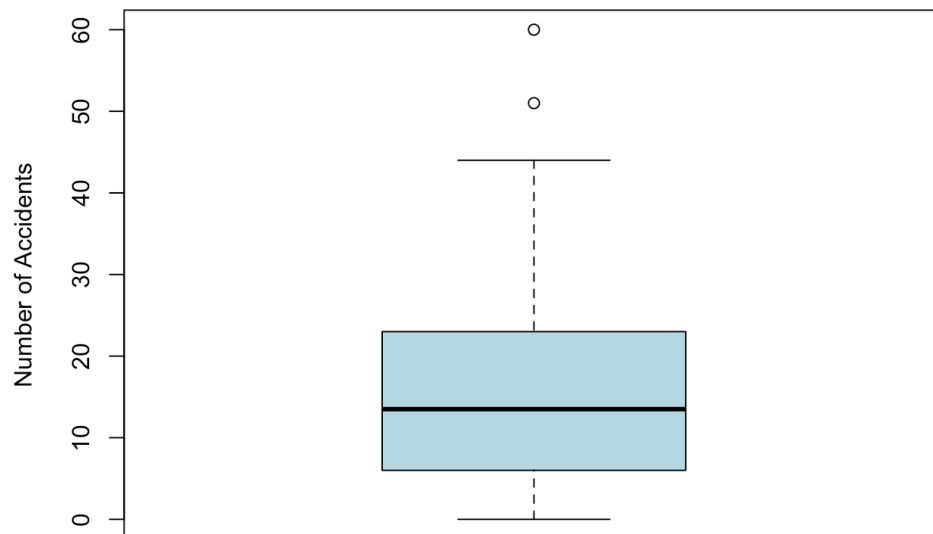
> range(filter(roadrace, Sex=="M")$Age)
[1]  9 83

> range(filter(roadrace, Sex=="F")$Age)
[1]  7 86
```

- The age distributions for male and female runners are similar in spread but differ somewhat in center.
- Male runners tend to be slightly older on average.
- IQR (Q3 – Q1):
 - Males = 51 – 30 = 21
 - Females = 46 – 28 = 18
- Both groups have comparable variability, though males show slightly greater spread by SD and IQR. Females have a slightly wider overall range due to more extreme values.

QUESTION 2

Boxplot of Motorcycle Accidents



Summary statistics :

```
> summary(motorcycle$`Fatal Motorcycle Accidents`)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.00   6.00   13.50   17.02   23.00   60.00
> sd(motorcycle$`Fatal Motorcycle Accidents`)
[1] 13.81256
> range(motorcycle$`Fatal Motorcycle Accidents`)
[1]  0 60
> IQR(motorcycle$`Fatal Motorcycle Accidents`)
[1] 17
```

Outliers :

```
> motorcycle[motorcycle$`Fatal Motorcycle Accidents` %in% boxplot.stats(motorcycle$`Fatal M
otorcycle Accidents`)$out, ]
      County Fatal Motorcycle Accidents
23 GREENVILLE          51
26   HORRY             60
```

Summary statistics analysis:

- The distribution of motorcycle fatalities is right-skewed because:
 - Mean (=17.02) > Median(=13.5)
- Most counties have relatively low to moderate numbers of fatalities, with the middle 50% of values lying between 6 and 23 accidents (given by 1st and 3rd quantile).
- Large standard deviation (13.81) indicates high variability across counties.
- Upper bound to calculate outliers : $Q3 + 1.5 \times IQR = 23 + 1.5(17) = 48.5$
Since both 51 and 60 are greater than 48.5, they are classified as outliers.
- Counties of outliers : Greenville (51) and Horry(60).

Reason for high no. of fatalities:

- Higher population density compared to other counties.
- Higher motorcycle usage.

QUESTION 3

A.

Here I use the differentiation technique to find MLE, where I maximize the log likelihood function $\log\{L(\theta)\}$.

$$f(x) = 0, x < 1 \text{ and } f(x) = \theta / x^{(\theta + 1)}, x \geq 1$$

$$\begin{aligned} L(\theta) &= \text{Product from } i=1 \text{ to } n (\theta / x_i^{(\theta + 1)}) \\ &= \theta^n * (\text{Product from } i=1 \text{ to } n (1 / x_i)) \end{aligned}$$

$$\log (L(\theta)) = n \ln(\theta) - (\theta + 1) * (\text{Summation from } i=1 \text{ to } n (\ln x_i))$$

Differentiating log likelihood, we get ,

$$dL/d\theta = n/\theta - (\text{Summation from } i=1 \text{ to } n (\ln x_i))$$

Equating the derivative to zero, we get,

$$\Rightarrow n/\hat{\theta} = \text{Summation from } i=1 \text{ to } n (\ln x_i)$$

$$\Rightarrow \theta_{\hat{}} = n / \text{Summation from } i=1 \text{ to } n (\ln x_i)$$

Hence the MLE of theta is

$$\Rightarrow \theta_{\hat{}} = n / \text{Summation from } i=1 \text{ to } n (\ln x_i)$$

B.

$$n=5$$

$$x_1 = 21.72, x_2 = 14.65, x_3 = 50.42, x_4 = 28.78, x_5 = 11.23$$

$$\text{Summation } i=1 \text{ to } 5 (\ln x_i) =$$

$$\ln(21.72) + \ln(14.65) + \ln(50.42) + \ln(28.78) + \ln(11.23)$$

$$\Rightarrow \text{Summation } i=1 \text{ to } 5 (\ln x_i) = 15.461$$

From MLE derived in part A,

$$\theta_{\hat{}} = n / \text{Summation}$$

$$= 5 / 15.461$$

$$= 0.323$$

```
> x <- c(21.72, 14.65, 50.42, 28.78, 11.23)
```

```
> n <- length(x)
```

```
> theta_hat <- n / sum(log(x))
```

```
> theta_hat
```

```
[1] 0.3233874
```

SECTION 2

QUESTION 1

A.Code to plot a barplot of runners in Road Race

```

17 # We create a barplot of the table Maine and label it using the function text
18 bp<- barplot(MaineOrNot, xlab="RunnerOrigin", ylab="Number of Runners")
19 text(bp, MaineOrNot, labels = MaineOrNot, pos=1)

```

B.Code to create 2 histograms for runners from Maine and Away

```

24 #Histogram for times from Away
25 h<- hist(roadrace$`Time (minutes)` [roadrace$Maine=="Maine"] ,
26         breaks = 7, main="Histogram for runners' times from Maine" ,
27         xlab = "" , col = "pink" )
28 # text function helps me label the histogram
29 text(h$mids, h$counts, labels = h$counts, pos = 3)
30
31 # Histogram for times from Maine
32 h<- hist(roadrace$`Time (minutes)` [roadrace$Maine=="Away"] ,
33         breaks = 7, main="Histogram for runners' times from Away" ,
34         xlab = "" , col = "yellow" )
35 # text function helps me label the histogram
36 text(h$mids, h$counts, labels = h$counts, pos = 3)
37

```

Code to generate summary statistics for runners from Maine and Away

```

38 # We filter for runners from Maine using filter function.
39 Maine_Runners <- filter(roadrace, Maine=="Maine")
40 #summary() helps us generate summary
41 summary(Maine_Runners$`Time (minutes)` )
42 # sd() and range() help calculate standard deviation and range.
43 sd(Maine_Runners$`Time (minutes)` )
44 range(Maine_Runners$`Time (minutes)` )
45
46 # We filter for runners from Away using filter() function
47 Away_Runners <- filter(roadrace, Maine=="Away")
48 # summary() helps us generate summary
49 summary(Away_Runners$`Time (minutes)` )
50 # sd() and range() help calculate standard deviation and range.
51 sd(Away_Runners$`Time (minutes)` )
52 range(Away_Runners$`Time (minutes)` )

```

C.Side by side boxplots of race time for Away and Maine runners

```

56 # Boxplot for runners' time in minutes
57 boxplot(`Time (minutes)` ~ Maine,
58         data = roadrace,
59         col = c("lightblue", "lightgreen"),
60         main = "Race Time Comparison",
61         ylab = "Time (minutes)")

```

D.Side by side boxplots for Age of Male and Female runners

```
63 # Boxplot for ages
64 boxplot(`Age` ~ Sex,
65         data = roadrace,
66         col = c("pink", "yellow"),
67         main = "Age Comparison",
68         ylab = "Age")
```

QUESTION 2

R code below shows the following :

1. Boxplot
2. Features of data distribution (Summary, Standard Deviation, Range, Inter-quartile range).
3. Outliers in boxplot

```
9 # Boxplot to show no. of motorcycle accidents in South Carolina
10 boxplot(motorcycle$`Fatal Motorcycle Accidents`,
11         main = "Boxplot of Motorcycle Accidents",
12         ylab = "Number of Accidents",
13         col = "lightblue")
14
15 # Features of data distribution
16 # Data summary , standard deviation, range and Inter-quartile range
17 summary(motorcycle$`Fatal Motorcycle Accidents`)
18 sd(motorcycle$`Fatal Motorcycle Accidents`)
19 range(motorcycle$`Fatal Motorcycle Accidents`)
20 IQR(motorcycle$`Fatal Motorcycle Accidents`)
21
22 # To find outliers in boxplot
23 motorcycle[motorcycle$`Fatal Motorcycle Accidents` %in%
24             boxplot.stats(motorcycle$`Fatal Motorcycle Accidents`)$out, ]
```

QUESTION 3

```
27 # Initializing the array as x
28 x <- c(21.72, 14.65, 50.42, 28.78, 11.23)
29 n <- length(x)
30 # Assigning the MLE expression to variable theta_hat
31 theta_hat <- n / sum(log(x))
32 # Printing result
33 theta_hat
```